Sector-length geometry of pure symmetric six-qubit states

A concise guide to the proved bounds and attained states

J. Martin O. Giraud E. Serrano-Ensástiga

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Scope. This short version collects the main results behind the interactive atlas. It gives the normalization, principal necessary inequalities, selected explicit states and boundary families, and the meaning of the attainment colors. Detailed proofs, finite exact and interval certificates, and the complete retained defining system are in the *full N=6 supplement* [1]. The displayed body is a *proved outer region*; the full achievable three-dimensional image is not yet characterized.

1 Coordinates, moments and two-copy weights

A normalized state is

$$|\psi\rangle = \sum_{k=0}^{6} c_k D_k, \qquad D_k = |D_6^{(k)}\rangle = |3, 3-k\rangle, \qquad \sum_k |c_k|^2 = 1.$$

For $\rho = |\psi\rangle\langle\psi|$, with $\sigma_0 = I$ and $\sigma_{1,2,3} = X, Y, Z$, define

$$S_k = \sum_{\mathrm{wt}(\mu)=k} \left[\mathrm{Tr}(\rho\, \sigma_{\mu_1} \otimes \cdots \otimes \sigma_{\mu_6})\right]^2, \qquad S_0 = 1. \tag{1}$$

Here $\mathrm{wt}(\mu)$ counts nonidentity factors. No binomial factor is divided out. Set $x = S_1$, $y = S_2$, $z = S_3$. Purity and symmetry eliminate the remaining sectors:

$$\boxed{S_4 = 45 + 10x - 2y - 3z, \quad S_5 = 3z - 9x, \quad S_6 = 18 - 2x + y - z.} \tag{2}$$

In particular $\sum_{k=0}^{6} S_k = 64$.

Let $J_i = \frac{1}{2}\sum_{a=1}^{6} \sigma_i^{(a)}$, $\boldsymbol{F} = \langle \boldsymbol{J} \rangle$ and $Q_{ij} = \langle \{J_i, J_j\}/2 \rangle - 4\delta_{ij}$. Then

$$f = |\boldsymbol{F}|, \quad \tau = \mathrm{Tr}\, Q^2, \qquad x = \frac{2}{3} f^2, \quad y = 5 + \frac{4}{15}\tau. \tag{3}$$

With orthonormal spin multipoles and squared norms a_ℓ, $x = 56a_1/3$ and $y = 5 + 168a_2/5$. Define

$$q = |\langle \mathsf{T}_3 \psi | \psi \rangle|^2 = 1 + \frac{x}{4} - \frac{z}{8}, \quad r = \sqrt{q}, \quad \Delta = 15 + 5x - y - z. \tag{4}$$

Time reversal acts as $\mathsf{T}_3 \sum_m c_m |3, m\rangle = \sum_m (-1)^{3-m} \overline{c_m} |3, -m\rangle$. For P_J the total-spin-J projector on two copies, put $p_J = \|P_J |\psi\rangle^{\otimes 2}\|^2$ $(J = 0, 2, 4, 6)$. Their exact values are

$$\begin{aligned} p_0 &= \frac{q}{7}, & p_2 &= \frac{5(y + z - 5x - 5)}{168}, \\ p_4 &= \frac{3(162 + 8x - 10y - 3z)}{616}, & p_6 &= \frac{401 + 137x + 35y + 5z}{1848}. \end{aligned} \tag{5}$$

They are nonnegative and sum to one. The singlet probability $p_0 = q/7$ must be distinguished from the overlap q.

2 The current outer body

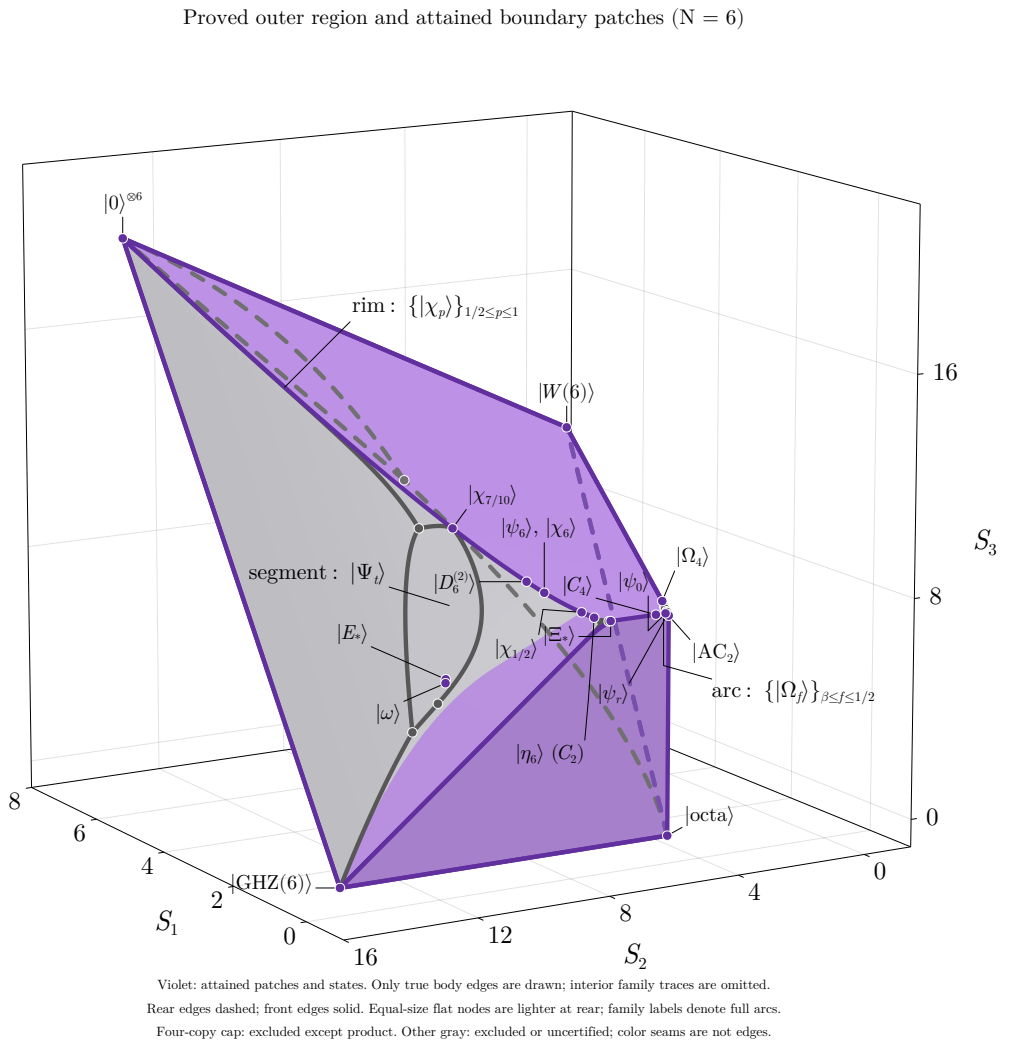


Figure 1. The current proved outer region $\mathcal{R}_4^{\mathrm{ref}}$, including the refined four-copy lower condition in Eq. (14). Snapshot: 6 November 2026.

Violet identifies constructively attained points, curves and surface patches. Each sampled state has normalized Dicke coefficients and independently recomputed sector lengths. A line segment is declared attained only when an actual state family covers it.

Gray means excluded or not yet certified. All 11 marked gray nodes are analytically excluded. This does not imply that every gray surface point has been excluded. The refined four-copy lower patch and its open crease arcs are an important exception: they are excluded except at the product point.

Rear edges are lighter and dashed. The mesh is a display approximation; color interpolation within a triangle is not an independent state construction. The interactive atlas supplies rotation, magnification, alternative and rear views, exact state expressions, sector tuples and Majorana constellations.

3 Principal necessary inequalities

The following compact collection records the main restrictions. It is not a replacement for the complete defining system: the figure also retains the stronger local support lifts and guarded flow bounds summarized in Section 7. All inequalities stated here are proved; no conjectural cut is assumed.

Positivity and the paired-purity face.

$$x \ge 0, \qquad y \ge 5, \qquad q \ge 0, \qquad \Delta \ge 0, \tag{6}$$

$$y + z \ge 5x + 5, \qquad z \ge 2x + \frac{12x}{4 + \sqrt{5(y-5)/2}}. \tag{7}$$

The first two statements are dipole and quadrupole positivity; the spin-two weight p_2 gives the first inequality in Eq. (7). The curved lower bound retains a three-vector Gram constraint together with the quadrupole norm. The condition $\Delta \ge 0$ is the paired-purity bound.

Fourth-sector envelope and the sharp supporting plane.

$$S_4 \ge \Psi(y), \qquad \boxed{65y + 63z - 217x \le 1011}, \tag{8}$$

$$\Psi(y) = \begin{cases} (6y-9)/7, & 5 \le y \le 99/10, \\ \left(14y - 5 - 160\sqrt{(y-5)/10}\right)/3, & 99/10 \le y \le 15, \\ 2y - 15, & y \ge 15. \end{cases} \tag{9}$$

The envelope follows from even four-qubit correlations and purity. The supporting plane follows from positive spin-projection blocks; in multipole notation its key input is $a_4 \ge 9q/286$. Its equality sector image is a segment, not the whole surrounding plane.

The full Gram supporting family. For every $t > 0$,

$$\boxed{10(y-5) + \frac{24x}{3+4t} + \frac{81q}{t} \ge \frac{1}{1+t}.} \tag{10}$$

Keeping the whole family gives a stronger boundary than selecting finitely many supporting planes. Its derivation uses pure-state rotation Gram matrices and Cauchy–Schwarz.

The finite-dipole implicit cap. Put $b = \sqrt{8 - \frac{8}{5}\sqrt{15}}$. Then

$$\boxed{\sqrt{y-5} \le b(1-r) + \sqrt{10}\, r + \sqrt{\frac{3}{2}}\sqrt{x}\sqrt{\frac{1-r}{1+r}}.} \tag{11}$$

This form retains the octupole compatibility improvement: the dipole coefficient is $\sqrt{3/2}$. The historical coefficient $14/5$ is weaker. The overlap has $0 \le r \le 1$ on pure states.

Useful consequences, not additional plotted surfaces, include $0 \le x \le 6$, $5 \le y \le 15$, $z \ge 10x/3$, $45/7 \le S_4 \le 35$, and $S_6 \ge 1 + (46/3)q$.

4 The retained spectral and four-copy refinements

Two complementary spectral remainders. Define

$$\delta = \sqrt{\Delta/40}, \qquad \alpha = \frac{1}{4} + \frac{\delta}{2}, \qquad \beta = \frac{1}{4} - \frac{\delta}{2},$$

$$H = \frac{1}{4} + \delta + 5\delta^2, \qquad K = (\sqrt{2}\,\alpha + 2\beta)^2 + 8\alpha\delta,$$

$$G = 85 + 15x - 5y - 6z + \sqrt{40\Delta}.$$

Pure-state compatibility gives

$$\boxed{G \geq \max\left\{\frac{4\Delta q}{K},\ \frac{16\Delta q}{\left(\sqrt{H + 4\alpha\sqrt{q}} + \sqrt{H}\right)^2}\right\}.} \tag{12}$$

Here Eqs. (6)–(7) give $0 \leq \Delta \leq 10$, so $\alpha, \beta \geq 0$. Both remainders are retained. Dropping the right-hand side gives the older roof $5y + 6z - 15x - 85 \leq \sqrt{40\Delta}$. The proof uses an auxiliary spin-two reduction with paired spectrum, a skew Takagi decomposition, overlap bounds and trace rearrangement.

Refined four-copy inequality. With the two-copy probabilities in Eq. (5),

$$\boxed{8p_2 + \frac{64}{81}p_4^2 \geq 2p_0p_6.} \tag{13}$$

Equivalently, in sector coordinates,

$$\boxed{\begin{aligned}&\frac{5}{21}(y + z - 5x - 5) + \left(\frac{162 + 8x - 10y - 3z}{231}\right)^2 \\ &\qquad \geq \frac{(8 + 2x - z)(401 + 137x + 35y + 5z)}{51744}.\end{aligned}} \tag{14}$$

The unrefined coefficient 1 in place of 64/81 is also proved, but contributes no extra boundary once Eq. (13) is imposed. The refined coefficient is not claimed optimal.

Equality and exclusion. Equality in either four-copy bound holds only for coherent products, up to collective rotation and global phase. Their sector tuple is $(x, y, z) = (6, 15, 20)$. Consequently the new exposed curved lower sheet, and its open crease arcs, do not represent attainable pure-state tuples. They remain valid necessary outer boundaries.

Proof mechanism. Positivity of a four-copy compression yields the original inequality. A projection onto total spin eight, followed by Haar-moment and variance estimates, supplies the 64/81 improvement. The full supplement provides the exact finite certificate and the equality classification. This summary does not replace those proof inputs.

5 Selected named states and sharp consequences

All representatives below are normalized; the remaining three sectors follow from Eq. (2). These are actual states, not fitted vertices.

State	Dicke representative	(S_1, S_2, S_3)
Product	D_0	$(6, 15, 20)$
GHZ(6)	$(D_0 + D_6)/\sqrt{2}$	$(0, 15, 0)$
Octahedral	$(D_1 + D_5)/\sqrt{2}$	$(0, 5, 0)$
$W(6)$	D_1	$(8/3, 5, 40/3)$
C_4	$(D_0 + \sqrt{3}D_4)/2$	$(0, 27/5, 8)$
Ω_4	$(\sqrt{3}D_0 + \sqrt{5}D_4)/\sqrt{8}$	$(1/6, 5, 25/3)$
AC_2	$(\sqrt{2}D_0 + \sqrt{5}D_3 + \sqrt{2}D_6)/3$	$(0, 5, 640/81)$
ψ_6	$(\sqrt{20}D_2 + D_5)/\sqrt{21}$	$(24, 405, 440)/49$
χ_6	$(2D_0 + \sqrt{3}D_5)/\sqrt{7}$	$(24, 405, 440)/49$
$\chi_{7/10}$	$(\sqrt{7}D_0 + \sqrt{3}D_5)/\sqrt{10}$	$(3/2, 99/10, 11)$
η_6	Eq. (16)	$(2/27, 65/9, 220/27)$

The two states ψ_6 and χ_6 have the same sector tuple, but belong to distinct collective-rotation orbits. Both attain the sharp minimum $S_4 = 45/7$. A sector tuple does not determine a state orbit.

A sharp excluded-vertex result. If $S_1 = 0$ and $S_2 = 5$, Eq. (10), multiplied by t and followed by $t \to \infty$, gives $81q \geq 1$. Using Eq. (4),

$$\boxed{S_1 = 0, \quad S_2 = 5 \quad \Longrightarrow \quad q \geq \frac{1}{81}, \qquad S_3 \leq \frac{640}{81}.} \tag{15}$$

The AC_2 state above attains equality. In particular the former outer vertex $(0, 5, 8)$ is excluded by a strict, sharp gap.

The sharp null paired-rim endpoint. An explicit state is

$$|\eta_6\rangle = \frac{5D_0 + \sqrt{35}D_2 + \sqrt{15}D_4 - \sqrt{21}D_6}{\sqrt{96}}. \tag{16}$$

The rim $q = 0$, $y = 7 + 3x$ is attainable exactly for $2/27 \leq x \leq 1/6$. The least possible x is $2/27$; its equality state is unique up to collective rotation and global phase.

6 Short explicit attained families

The following curves have exact sector maps. Each is a useful check of the geometry; none requires convex-hull interpolation.

The C_5 rim. For $1/2 \le p \le 1$,

$$|\chi_p\rangle = \sqrt{p}D_0 + \sqrt{1-p}D_5, \tag{17}$$

$$(x, y, z) = \left(\frac{2}{3}(5p-2)^2,\ 5 + 10p^2,\ 8 + \frac{4}{3}(5p-2)^2\right). \tag{18}$$

The whole rim has $q = 0$ and saturates the paired-spectral roof; the fourth-sector envelope also saturates for $p \ge 7/10$.

A C_4 null-roof curve. For $b_c \le f \le 1/2$,

$$|\Omega_f\rangle = \frac{\sqrt{1+f}D_0 + \sqrt{3-f}D_4}{2}, \tag{19}$$

$$(x, y, z) = \left(\frac{2}{3}f^2,\ 5 + \frac{2}{5}(1-2f)^2,\ 8 + \frac{4}{3}f^2\right). \tag{20}$$

The threshold $b_c = 0.366379159399607\ldots$ is the unique root in $(0, 1/2)$ of $16b_c^4 - 78b_c^3 + 92b_c^2 + 6b_c - 11 = 0$. The familiar interval $3/8 \le f \le 1/2$ is a proper subset.

A C_3 zero-dipole boundary curve. For $0 \le r \le 1/9$,

$$|\psi_r\rangle = \frac{\sqrt{1-r}}{2}(D_0 + D_6) + \sqrt{\frac{1+r}{2}}D_3, \tag{21}$$

$$(x, y, z) = \left(0,\ 5 + \frac{(1-9r)^2}{10},\ 8(1-r^2)\right). \tag{22}$$

It joins ψ_0 to AC_2. It is the lower-y side curve, not the whole zero-dipole boundary.

The filtered-octahedral C_3 trace. For $0 \le t \le 1$, let

$$A_t = \sqrt{\frac{4 + 5t + 3\sqrt{t(t+8)}}{18}}, \quad B_t = \frac{\sqrt{5(1-t)}}{3}, \quad C_t = \sqrt{\frac{4 + 5t - 3\sqrt{t(t+8)}}{18}}.$$

Then

$$|F_t\rangle = A_t D_0 + B_t D_3 - C_t D_6, \tag{23}$$

$$(x, y, z) = \left(\frac{2}{3}t(t+8),\ 5 + 10t^2,\ \frac{20}{3}t(4-t)\right). \tag{24}$$

This trace lies on $y + z = 5x + 5$ and joins collectively rotated octahedral and product representatives. It bounds a constructively filled patch.

7 Boundary coverage, limitations and provenance

Attained faces. The full supplement determines the complete pure-state images on the lower spin-two plane $y + z = 5x + 5$, the quadrupole-free face $y = 5$, and the paired-purity plane $\Delta = 0$. Their attainable subsets are not identified with the whole intersection of an arbitrary outer polytope. Cyclic C_2–C_6 families, vertex-state superpositions and collective filters supply explicit constructions.

The rank-four equality segment. All attainable tuples saturating the sharp plane in Eq. (8) lie on, and fill, the segment

$$(x, y, z) = \left(\frac{3}{2} - \frac{474}{169}q,\ \frac{99}{10} + \frac{3234}{845}q,\ 11 - \frac{2300}{169}q\right), \quad 0 \le q \le \frac{169}{484}. \tag{25}$$

Its endpoints are $\chi_{7/10}$ and

$$|\omega\rangle = \frac{(\sqrt{21} + 3\sqrt{2})D_1 + 7\sqrt{2}D_3 + (\sqrt{21} - 3\sqrt{2})D_5}{4\sqrt{11}}, \tag{26}$$

with tuple $(63/121, 618/55, 756/121)$. The complete state-orbit classification includes chiral branches and an exceptional C_2 orbit E_* in the interior of the same sector segment. The exceptional orbit adds no new sector face; its explicit coefficients are available in the atlas and full supplement.

Sharper compatibility cuts. The actual figure also retains nine exact null-state supporting estimates, their positive-overlap lifts, a C_4 threshold lift, and a guarded zero-dipole flow comparison. These can improve very small neighborhoods. They are not reproduced as a large coefficient appendix here; use the complete defining system in the full supplement to reproduce the outer body exactly. Rounded anchor values are not substituted for its exact rational and algebraic inputs.

What remains open. Complete complex-C_2 null-roof bands and four certified unrestricted strips are attained. Restricted-family optimization and local Hessian stability do not establish unrestricted global sharpness outside those certified ranges. The zero-dipole side is sharp on stated ranges; two tiny residual tails remain explicitly bracketed. Thus the full three-dimensional pure-state sector image remains unresolved. Some global-sharpness and continuation steps use exact or directed-interval computer-assisted certificates; failed numerical fitting is never an exclusion proof.

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References

[1] *Proved outer bounds for the sector lengths of pure symmetric six-qubit states*, full supplement, updated 6 November 2026. Separate extended technical document; not distributed on this site.

[2] *Sector-length sets: pure symmetric states, $N = 3$–6.* Interactive atlas and reproducible state data.

[3] *Multiqubit monogamy relations beyond shadow inequalities*, Phys. Rev. A **113**, 012415 (2026). doi:10.1103/PRA.113.012415.